



Handling Decimal Precision

Practical Skills Module 2: Topic 4

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Topic Code	PM-02-PS04:
Topic	Use double data types in Python

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0401 Use double data types in Python

Applied Knowledge

- AK0401 Concept, definition and functions of Python data types

Internal Assessment Criteria

- IAC0401 Expected results with Python double are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-02



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario: Banking System - Interest Calculation for Savings Accounts

You are tasked with creating a Python program for a banking system that calculates interest for customers' savings accounts. In this scenario, you use **double-like data types** (represented as float in Python) to handle financial data such as balances, interest rates, and monthly interest.

Create a Python file called **banking_system.py** and complete the following steps:

Step 1: Assign float values to customer balances.

- Create **three customer account balances**.
- Use float values to represent each balance (e.g., `10000.50`), including cents.
- Store each balance in its own variable with a clear name.

Tip: These represent the amount each customer has in their savings account.

Step 2: Set the interest rate.

- Assign a float value to a variable to represent the **annual interest rate** (e.g., `3.5` for 3.5%).

Note: This is an annual rate, but you will calculate monthly interest in the next step.

Step 3: Calculate monthly interest.

- For each customer:
 1. Convert the annual rate to a monthly rate by dividing it by 12.
 2. Calculate the monthly interest using the formula:

```
monthly_interest = account_balance × (monthly_interest_rate / 100)
```

Example: If the balance is `10000.50` and the annual interest rate is `3.5`, the monthly interest rate would be $3.5 / 12 = 0.291666\dots$

Then: `monthly_interest = 10000.50 × (0.291666 / 100)`

Step 4: Update the account balance.

- Add the calculated monthly interest to the customer's existing account balance.
- Update the original balance variable to reflect the new total.

Step 5: Format the result for display.

- Print each customer's updated account balance.
- Format the output to **two decimal places** for clarity and professionalism.

- *Example output:*

Customer 1's updated balance: R10029.69

Customer 2's updated balance: R8450.13



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
